

### Proof of the Exponential Decay Property of the $Q$ -function

This section establishes the exponential decay property of each agent's local action-value function. The result is formalized in the following proposition.

**Proposition:** Let  $\mathcal{N}=\{1, \dots, n\}$  denotes the set of agents. For each agent  $i$ , let  $\mathcal{N}_i^\kappa$  denotes its  $\kappa$ -hop neighborhood, including agent  $i$ , and let  $\mathcal{N}_{-i}^\kappa = \mathcal{N}/\mathcal{N}_i^\kappa$  denotes the set of agents outside the  $\kappa$ -hop neighborhood. Accordingly, the global state and action are partitioned as  $s=(s_{\mathcal{N}_i^\kappa}, s_{\mathcal{N}_{-i}^\kappa})$ ,  $a=(a_{\mathcal{N}_i^\kappa}, a_{\mathcal{N}_{-i}^\kappa})$ .

Fixing a tuple of localized policy parameters  $\theta=(\theta_1, \dots, \theta_n)$ , the global action-value function is defined as:

$$\begin{aligned} Q(s, a) &= \mathbb{E}_{a^t \sim \pi^\theta(\cdot|s^t)} \left[ \sum_{t=0}^{\infty} \gamma^t r(s^t, a^t) \mid s^0 = s, a^0 = a \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{a_i^t \sim \pi^{\theta_i}(\cdot|s_i^t)} \left[ \sum_{t=0}^{\infty} \gamma^t r_i(s_i^t, a_i^t) \mid s^0 = s \right] = \frac{1}{n} \sum_{i=1}^n Q_i(s, a) \end{aligned} \quad (\text{A-1})$$

where  $Q_i(s, a)$  denotes the  $Q$  function for the individual reward  $r_i$ .

Suppose that the reward span of each agent is uniformly bounded. Specifically, there exists a finite constant  $r_{\max} > 0$  such that:

$$\sup_{s_i, a_i} r_i(s_i, a_i) - \inf_{s_i, a_i} r_i(s_i, a_i) \leq r_{\max}, \forall i \in \mathcal{N} \quad (\text{A-2})$$

Then, for any  $i \in \mathcal{N}$ ,  $s_{\mathcal{N}_i^\kappa} \in \mathcal{S}_{\mathcal{N}_i^\kappa}$ ,  $s_{\mathcal{N}_{-i}^\kappa}, s'_{\mathcal{N}_{-i}^\kappa} \in \mathcal{S}_{\mathcal{N}_{-i}^\kappa}$ ,  $a_{\mathcal{N}_i^\kappa} \in \mathcal{A}_{\mathcal{N}_i^\kappa}$ ,  $a_{\mathcal{N}_{-i}^\kappa}, a'_{\mathcal{N}_{-i}^\kappa} \in \mathcal{A}_{\mathcal{N}_{-i}^\kappa}$ , the following inequality holds:

$$\left| Q_i(s_{\mathcal{N}_i^\kappa}, s_{\mathcal{N}_{-i}^\kappa}, a_{\mathcal{N}_i^\kappa}, a_{\mathcal{N}_{-i}^\kappa}) - Q_i(s_{\mathcal{N}_i^\kappa}, s'_{\mathcal{N}_{-i}^\kappa}, a_{\mathcal{N}_i^\kappa}, a'_{\mathcal{N}_{-i}^\kappa}) \right| \leq \frac{r_{\max}}{1-\gamma} \gamma^\kappa \quad (\text{A-3})$$

**Proof:** By the definition of  $Q_i$  and the triangle inequality, we obtain:

$$\begin{aligned} &\left| Q_i(s_{\mathcal{N}_i^\kappa}, s_{\mathcal{N}_{-i}^\kappa}, a_{\mathcal{N}_i^\kappa}, a_{\mathcal{N}_{-i}^\kappa}) - Q_i(s_{\mathcal{N}_i^\kappa}, s'_{\mathcal{N}_{-i}^\kappa}, a_{\mathcal{N}_i^\kappa}, a'_{\mathcal{N}_{-i}^\kappa}) \right| \\ &\leq \sum_{t=0}^{\infty} \left| \mathbb{E} \left[ \gamma^t r_i(s_i^t, a_i^t) \mid s^0 = (s_{\mathcal{N}_i^\kappa}, s_{\mathcal{N}_{-i}^\kappa}), a^0 = (a_{\mathcal{N}_i^\kappa}, a_{\mathcal{N}_{-i}^\kappa}) \right] - \mathbb{E} \left[ \gamma^t r_i(s_i^t, a_i^t) \mid s^0 = (s_{\mathcal{N}_i^\kappa}, s'_{\mathcal{N}_{-i}^\kappa}), a^0 = (a_{\mathcal{N}_i^\kappa}, a'_{\mathcal{N}_{-i}^\kappa}) \right] \right| \end{aligned} \quad (\text{A-4})$$

Denote  $s=(s_{\mathcal{N}_i^\kappa}, s_{\mathcal{N}_{-i}^\kappa})$ ,  $a=(a_{\mathcal{N}_i^\kappa}, a_{\mathcal{N}_{-i}^\kappa})$ ,  $s'=(s_{\mathcal{N}_i^\kappa}, s'_{\mathcal{N}_{-i}^\kappa})$ ,  $a'=(a_{\mathcal{N}_i^\kappa}, a'_{\mathcal{N}_{-i}^\kappa})$ . Let  $\mu_i^t$  be the distribution of  $(s_i^t, a_i^t)$  conditioned on  $(s^0, a^0)=(s, a)$  under the joint policy  $\theta$ . Similarly, let  $\mu_i'^t$  denote the distribution of  $(s_i^t, a_i^t)$  conditioned on  $(s^0, a^0)=(s', a')$  under the joint policy  $\theta$ . For any subset  $\mathcal{J} \subseteq \mathcal{N}$ , define:

$$s_{\mathcal{J}}^t := (s_j^t)_{j \in \mathcal{J}}, a_{\mathcal{J}}^t := (a_j^t)_{j \in \mathcal{J}} \quad (\text{A-5})$$

We next show that:

$$\mu_{t,i} = \mu_{t,i}', 0 \leq t \leq \kappa \quad (\text{A-6})$$

To prove (A-6), define the shrinking neighborhood:

$$\mathcal{H}_t := \mathcal{N}_i^{\kappa-t}, t = \{0, \dots, \kappa\} \quad (\text{A-7})$$

Let

$$v_t = \text{Law}_{P, \pi^\theta} \left( s_{\mathcal{H}_t}^t, a_{\mathcal{H}_t}^t \mid s^0 = s, a^0 = a \right), v_t' = \text{Law}_{P, \pi^\theta} \left( s_{\mathcal{H}_t}^t, a_{\mathcal{H}_t}^t \mid s^0 = s', a^0 = a' \right) \quad (\text{A-8})$$

Here,  $\text{Law}_{P, \pi^\theta}$  denotes the conditional probability distribution induced by the transition kernel  $P$  and the joint policy  $\pi^\theta$ . Thus,  $v_t$  and  $v_t'$  are the conditional joint distributions of the state-action variables within  $\mathcal{H}_t$  under the two initial state-action pairs, respectively. We first establish the stronger result:

$$v_t = v_t', 0 \leq t \leq \kappa \quad (\text{A-9})$$

The proof proceeds by induction on  $t$ . For  $t=0$ ,  $\mathcal{H}_0 = \mathcal{N}_i^\kappa$ . By construction, the two initial state-action pairs coincide on  $\mathcal{N}_i^\kappa$ :

$$s_{\mathcal{H}_0} = s_{\mathcal{N}_i^\kappa} = s'_{\mathcal{H}_0}, a_{\mathcal{H}_0} = a_{\mathcal{N}_i^\kappa} = a'_{\mathcal{H}_0} \quad (\text{A-10})$$

Hence, under the respective conditioning events,

$$v_0 = \delta_{(s_{\mathcal{H}_0}, a_{\mathcal{H}_0})} = \delta_{(s'_{\mathcal{H}_0}, a'_{\mathcal{H}_0})} = v_0' \quad (\text{A-11})$$

where  $\delta_x$  denotes the Dirac probability measure concentrated at  $x$ .

Suppose that  $v_t = v_t'$  holds for some  $t < \kappa$ . Consider any agent:

$$j \in \mathcal{H}_{t+1} = \mathcal{N}_i^{\kappa-t-1} \quad (\text{A-12})$$

For every  $l \in \mathcal{N}_j$ , the triangle inequality for graph distances gives:

$$d(i, l) \leq d(i, j) + d(j, l) \leq \kappa - t - 1 + 1 = \kappa - t \quad (\text{A-13})$$

Therefore,

$$\mathcal{N}_j \subseteq \mathcal{N}_i^{\kappa-t} = \mathcal{H}_t \quad (\text{A-14})$$

According to the local transition structure:

$$P(s^{t+1} \mid s^t, a^t) = \prod_{m=1}^n P_m(s_m^{t+1} \mid s_{\mathcal{N}_m}^t, a_m^t) \quad (\text{A-15})$$

In addition, the localized joint policy satisfies:

$$\pi^\theta(a^t|s^t) = \prod_{m=1}^n \pi^{\theta_m}(a_m^t|s_m^t) \quad (\text{A-16})$$

For any  $x \in \mathcal{S}_{\mathcal{H}_t}, u \in \mathcal{A}_{\mathcal{H}_t}, x' \in \mathcal{S}_{\mathcal{H}_{t+1}}, u' \in \mathcal{A}_{\mathcal{H}_{t+1}}$ , define the state-action transition kernel:

$$K_t(x', u' | x, u) = \prod_{j \in \mathcal{H}_{t+1}} P_j(x'_j | x_{\mathcal{N}_j}, u_j) \pi_j(u'_j | x'_j) \quad (\text{A-17})$$

The inclusion in (A-13) guarantees that  $x_{\mathcal{N}_j}$  is contained in  $x$  for every  $j \in \mathcal{H}_{t+1}$ . Moreover,  $\mathcal{H}_{t+1} \subseteq \mathcal{H}_t$ , so each required action  $u_j$  is also contained in  $u$ . Thus,  $K_t$  is fully determined by the state-action variables in  $\mathcal{H}_t$ . It is identical under the two initial conditions because both stochastic processes follow the same transition kernel  $P$  and the same policy  $\pi^\theta$ .

Since the state and action spaces are finite, for every:

$$(x', u') \in \mathcal{S}_{\mathcal{H}_{t+1}} \times \mathcal{A}_{\mathcal{H}_{t+1}} \quad (\text{A-18})$$

we have:

$$\begin{aligned} v_{t+1}(x', u') &= \sum_{x \in \mathcal{S}_{\mathcal{H}_t}} \sum_{u \in \mathcal{A}_{\mathcal{H}_t}} v_t(x, u) K_t(x', u' | x, u) \\ &= \sum_{x \in \mathcal{S}_{\mathcal{H}_t}} \sum_{u \in \mathcal{A}_{\mathcal{H}_t}} v'_t(x, u) K_t(x', u' | x, u) \\ &= v'_{t+1}(x', u') \end{aligned} \quad (\text{A-19})$$

The second equality follows directly from the induction hypothesis:  $v_t = v'_t$ .

Hence:

$$v_{t+1} = v'_{t+1} \quad (\text{A-20})$$

Together with the initial equality in (A-10), mathematical induction yields

$$v_t = v'_t, 0 \leq t \leq \kappa \quad (\text{A-21})$$

Finally,  $i \in \mathcal{H}_t$  for every  $0 \leq t \leq \kappa$ . Therefore,  $\mu'_t$  and  $\mu''_t$  are the corresponding marginal distributions of  $v_t$  and  $v'_t$ . For any subset  $B \subseteq \mathcal{S}_i \times \mathcal{A}_i$ ,

$$\begin{aligned} \mu_{t,i}(B) &= v_t(\{(x, u) : (x_i, u_i) \in B\}) \\ &= v'_t(\{(x, u) : (x_i, u_i) \in B\}) \\ &= \mu'_{t,i}(B) \end{aligned} \quad (\text{A-22})$$

Thus,

$$\mu_{t,i} = \mu'_{t,i}, 0 \leq t \leq \kappa \quad (\text{A-23})$$

which proves (A-6).

By (A-6), the two conditional distributions coincide for all  $0 \leq t \leq \kappa$ . Therefore,

$$\mathbb{E}_{(s'_i, a'_i) \sim \mu'_i} r_i(s'_i, a'_i) = \mathbb{E}_{(s'_i, a'_i) \sim \mu''_i} r_i(s'_i, a'_i), 0 \leq t \leq \kappa \quad (\text{A-24})$$

It follows from (A-4) that:

$$\begin{aligned} &\left| Q_i(s_{\mathcal{N}_i^\kappa}, s_{\mathcal{N}_{-i}^\kappa}, a_{\mathcal{N}_i^\kappa}, a_{\mathcal{N}_{-i}^\kappa}) - Q_i(s_{\mathcal{N}_i^\kappa}, s'_{\mathcal{N}_{-i}^\kappa}, a_{\mathcal{N}_i^\kappa}, a'_{\mathcal{N}_{-i}^\kappa}) \right| \\ &\leq \sum_{t=0}^{\infty} \left| \mathbb{E}[\gamma^t r_i(s'_i, a'_i)] - \mathbb{E}[\gamma^t r_i(s'_i, a'_i)] \right| \\ &= \sum_{t=\kappa+1}^{\infty} \gamma^t \left| \mathbb{E}_{(s'_i, a'_i) \sim \mu'_i} r_i(s'_i, a'_i) - \mathbb{E}_{(s'_i, a'_i) \sim \mu''_i} r_i(s'_i, a'_i) \right| \end{aligned} \quad (\text{A-25})$$

For any bounded function  $f$  and probability distributions  $\mu$  and  $\nu$ ,

$$\left| \mathbb{E}_\mu[f] - \mathbb{E}_\nu[f] \right| \leq \text{span}(f) \text{TV}(\mu, \nu) \quad (\text{A-26})$$

where  $\text{span}(f) := \sup f - \inf f$ , and  $\text{TV}(\mu, \nu) := \sup_B |\mu(B) - \nu(B)|$  denotes the total variation distance.

Applying (A-24) to the local reward function gives:

$$\begin{aligned} &\left| \mathbb{E}_{(s'_i, a'_i) \sim \mu'_i} r_i(s'_i, a'_i) - \mathbb{E}_{(s'_i, a'_i) \sim \mu''_i} r_i(s'_i, a'_i) \right| \\ &\leq \text{span}(r_i) \text{TV}(\mu'_i, \mu''_i) \\ &\leq r_{\max} \text{TV}(\mu'_i, \mu''_i) \end{aligned} \quad (\text{A-27})$$

where the last inequality follows from the uniform reward-span bound in (A-2). Consequently,

$$\begin{aligned}
& \left| \mathcal{Q}_i(s_{N_i^\kappa}, s_{N_{-i}^\kappa}, a_{N_i^\kappa}, a_{N_{-i}^\kappa}) - \mathcal{Q}_i(s_{N_i^\kappa}, s'_{N_{-i}^\kappa}, a_{N_i^\kappa}, a'_{N_{-i}^\kappa}) \right| \\
&= \sum_{l=\kappa+1}^{\infty} \gamma^l r_{\max} \text{TV}(\mu'_i, \mu''_i) \\
&\leq r_{\max} \sum_{l=\kappa+1}^{\infty} \gamma^l \\
&= \frac{r_{\max}}{1-\gamma} \gamma^{\kappa+1}
\end{aligned} \tag{A-28}$$

where the second inequality follows from:

$$\text{TV}(\mu'_i, \mu''_i) \leq 1 \tag{A-29}$$

and the last inequality follows from  $0 < \gamma < 1$ . Hence, the influence of the state-action variables outside the  $\kappa$ -hop neighborhood on the local action-value function decays exponentially with  $\kappa$ . This completes the proof.